Assignment 03: Application Deployment and Scaling on Kubernetes

The objective of this assignment is to deepen your understanding of Kubernetes and its deployment and scaling capabilities. You will gain hands-on experience with Kubernetes by deploying and managing containerized applications, configuring scaling options, and exploring various Kubernetes features.

Milestone 01: Kubernetes Installation and Setup

- Install a local Kubernetes cluster using **Minikube**
- Validate the cluster: You can validate the cluster by checking the available nodes of the cluster

The deliverable of this milestone is to provide the screenshots of the terminal and dashboard, which shows the installation and running of Minikube.

Milestone 02: Deploying and Managing Applications

1. Use the **pucitcloudnet/fft** [1] docker image as a sample application for deployment. The following images contain an application code and Apache server configuration. Figure 1 shows the output of the application available in the **pucitcloudnet/fft** docker image.
2. Create Kubernetes deployment manifests: Write YAML files that define the desired state of your application, including the container image, ports, and other necessary configurations.
3. Deploy the application: Use the kubectl command to deploy your application to the Kubernetes cluster.
4. Verify the application: Confirm that your application is running correctly by checking the pod status and accessing the deployed service.

The deliverable of this milestone is to provide the screenshots of the terminal, Minikube dashboard and deployed application accessing from the browser.

Milestone 03: Scaling and Load Balancing

- Configure horizontal pod autoscaling (HPA): Write an HPA manifest that defines the scaling rules based on CPU or custom metrics.
- Test the autoscaling behavior: Generate load on your application using HTTPerf [2] to trigger scaling and observe the increase or decrease in the number of pod replicas.
- Implement service and load balancing: Create a Kubernetes Service and expose it using a LoadBalancer to distribute traffic to your application.

The deliverable of this milestone is to provide the screenshots of the terminal, Minikube dashboard, deployed services, and httperf script output.

Submission Guidelines

You need to prepare a small report in PDF format containing deliverables for all the milestones before **November 09, 2026, 11:59 pm via Google Classroom**. Also, submit a loom video URL (maximum 4 minutes) showing all the relevant steps and working.

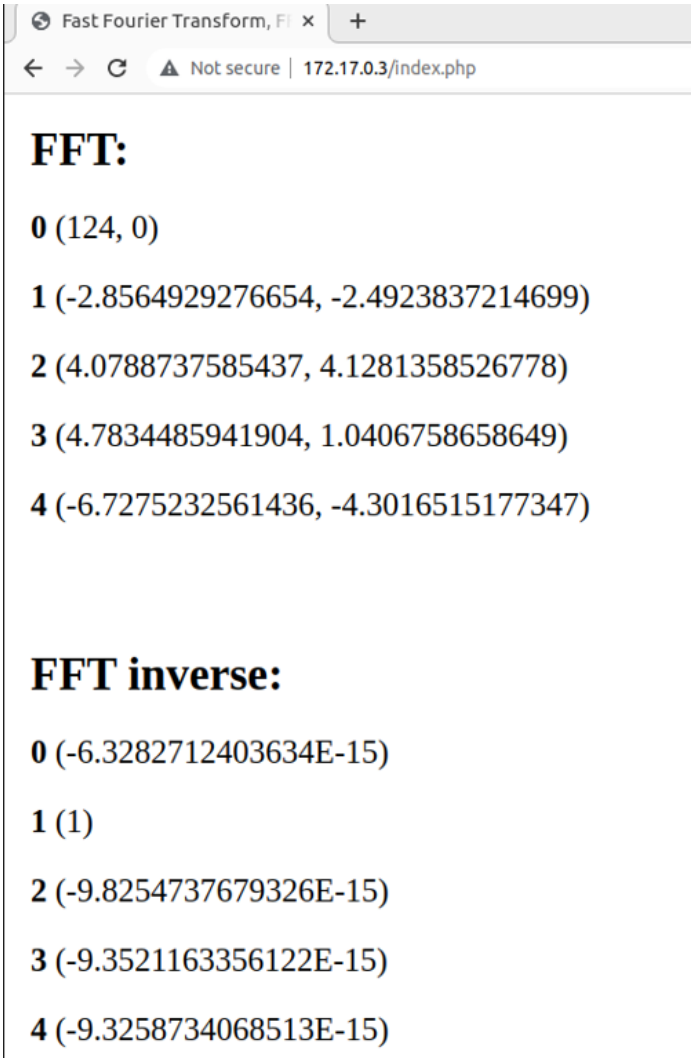


Figure 1: FFT Application

References

[1] pucitcloudnet/fft https://hub.docker.com/repository/docker/pucitcloudnet/fft/general

[2] httperf: A Tool for Measuring Web Server Performance. *http://www.labs.hpe.com/research/linux/httperf/docs.php*.